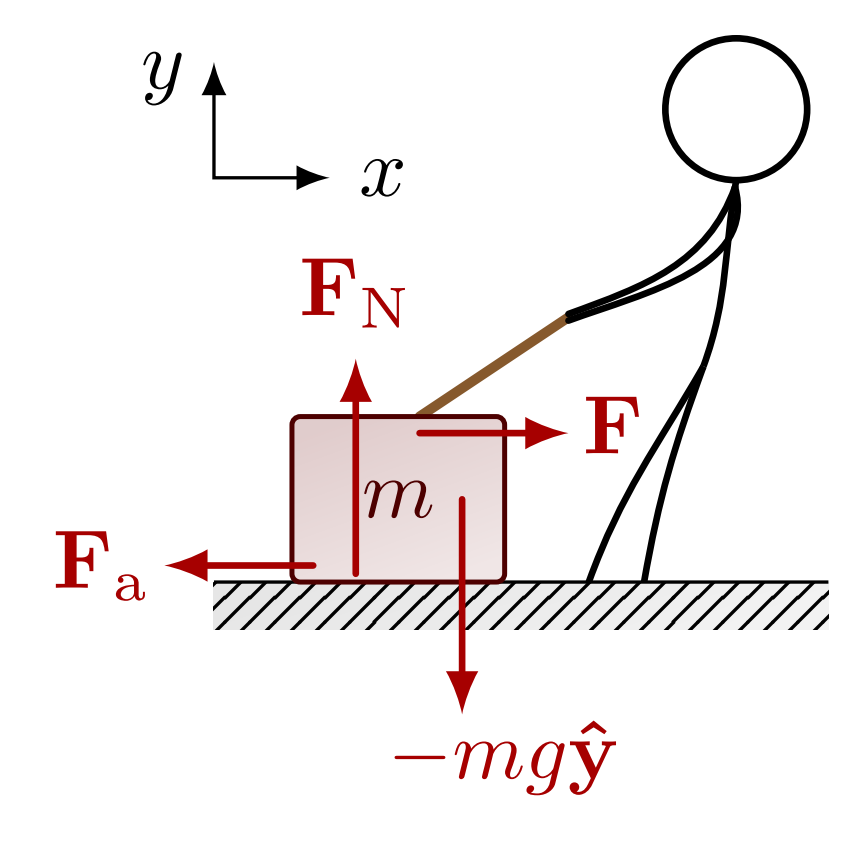
y
x
$\mathbf{F}_\mathrm{N}$
$\mathbf{F}$
m
$\mathbf{F}_\mathrm{a}$
$-mg\hat{\mathbf{y}}$